

- AMD DEVELOPER HACKATHON ACT II
- ACT I — 1ST PLACE AI AGENTS
- TRACK 3 UNICORN
- GEMMA 27B · FIREWORKS AI
- AMD RADEON → MI300X

REPO

MIND

v3

The on-premises coding agent for the engineers who **can't touch cloud AI** — a real coding model + 256K context on a **single AMD GPU**, air-gapped, at a cost that **pencils out for the enterprise.**

192GB AMD MI300X HBM3	256K token context (vLLM)	\$4.12 ACT I verified benchmark	39 CUDA→HIP rules + kernel launch
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Every tool that makes developers productive just got **banned**.

JP MORGAN CHASE

50,000

DEVELOPERS · RESTRICTED CLOUD AI CODING TOOLS¹

~50,000 developers, restricted from cloud AI coding. McKinsey's ~37% is a task-level coding speedup — applied to the coding share of dev time, recoverable value is on the order of **\$1–2B/yr**, not 37% of payroll.

≈ **\$1–2B/yr recoverable¹**

APPLE INC.

30,000+

ENGINEERS · RESTRICTED EXTERNAL AI CODING¹

GitHub Copilot and Google Gemini blocked. IP exfiltration risk: one leak event = \$100M–\$1B liability exposure. All AI coding tools blocked company-wide.

IP leak surface: unbounded

illustrative¹

GOLDMAN SACHS + EU ENTERPRISES

€35M

EU AI ACT · TIERED PENALTIES² · PHASING IN 2026

Goldman leadership has spoken publicly about immature AI governance frameworks.¹ EU AI Act obligations phase in through 2026 — up to €35M for the most serious violations.

Clock is ticking

Illustrative of a real segment: large regulated employers that restricted cloud AI coding. **REPOMIND is built for this gap** — these companies are not represented as customers.

¹ Per public reporting (press coverage / public executive statements). Figures are illustrative estimates, not company-confirmed.

Each judge. Each lens. Each **specific angle**.

N

Nick Ni · AMD VP Developer Ecosystem
Hardware platform, ROCm adoption, developer ecosystem

Your angle: The CUDA→ROCm migration agent directly attacks NVIDIA's ecosystem lock-in — AMD's #1 adoption barrier. 50M+ lines of CUDA on GitHub, waiting for exactly this tool. Every migration = AMD MI300X adoption. The hardware slide is built around YOUR hardware math.

I

Ian Ballantyne · Google DeepMind
AI developer ecosystem, Gemma integration

Relevant angle: Gemma 27B is load-bearing, not a cheap tier: on deep queries it triages the file manifest, selects the relevant subset, and hands the MI300X a focused context — a measured token reduction surfaced live. Gemma does the context-engineering the big model would otherwise waste 256K compute on (smart-routing mode; off under air-gap).

S

Suneeth Maraboina · Apple Technology Leader
Enterprise engineering, large-scale systems

Relevant angle: Slide 2 maps to the constraint large regulated engineering orgs face — restricted external AI coding tools. REPOMIND's enforced air-gap mode (AIR_GAP=1, egress hard-refused) and path-sandboxed tools are designed for the on-prem posture such infosec reviews require. Not a substitute for a full security review — a credible starting architecture.

P

Pawel Czech · lablab.ai — ACT I Judge
Hackathon evaluation, continuity narrative

Your angle: ACT I 1st Place was your call. AMD published an official testimonial. v3 adds Docker (required), Gemma routing (+\$2K prize), CUDA→ROCm, and multi-repo support — every ACT II evaluation criterion, addressed. The trajectory from v1 to v3 is visible and verified.

R

Rajat Gupta · Netflix ML Platform
Large-scale ML infrastructure, CUDA codebases

Your angle: Netflix operates CUDA-heavy ML pipelines. REPOMIND's agent can analyze any codebase and generate ROCm migration patches — enabling Netflix to evaluate AMD hardware as an alternative to NVIDIA at the infrastructure level, with automated migration instead of manual porting.

A

Andrei Kozlov · ENDGAME Ventures
Investment thesis, market, moat

Relevant angle: An underserved on-prem segment with a regulatory tailwind (EU AI Act phasing in through 2026, tiered penalties²). Incumbents exist but none are AMD-native. The wedge is first-mover, not a permanent moat. Pricing on slide 14: \$500–\$15K/month tiers + usage.

Four dimensions of pain. One solution vector.

COMPANY	BANNED TOOLS	SCALE	COMPLIANCE TRIGGER	ANNUAL RISK	REPOMIND V3
JP Morgan Chase NYSE: JPM · Financial Services	CURSOR ALL CLOUD AI	50,000 developers	MAS TRM · Basel III · SOX · SEC Rule 17a-4	~\$1–2B recoverable/yr ¹	✓ ON-PREM DOCKER
Apple Inc. NASDAQ: AAPL · Consumer Electronics	COPILOT GEMINI	30,000+ engineers	Trade secrets · IP licensing risk · Secrecy NDA policy	\$1B+ per IP incident	✓ ZERO EGRESS*
Goldman Sachs NYSE: GS · Investment Banking	ALL EXTERNAL AI	40,000 employees	SEC · CFTC · FINRA · Public statements on immature AI governance ¹	≤€35M top-tier max ²	✓ HEURISTIC PRE-SCAN
EU Enterprises GDPR jurisdiction · Aug 2026 enforcement	US-HOSTED AI	Millions affected devs	EU AI Act Art. 12 · GDPR data residency · tiered penalties ²	tiered ² €7.5M–€35M	✓ ON-PREMISES

The real market: enterprises that restricted cloud AI coding and need on-prem. Incumbents exist (Tabnine, Cody Enterprise) — **none combine air-gap + MI300X 256K + CUDA→ROCm.**

^{*} Air-gap mode: AIR_GAP=1 — Fireworks egress hard-refused at the client; all queries on-prem MI300X.

² EU AI Act penalties are tiered: ≤€7.5M/1% (Art.12-type record-keeping) up to €35M/7% (Art.5 prohibited practices). Not a per-coding-tool fine.

COMPETITIVE LANDSCAPE

No competitor combines the full on-prem stack.

Grid compares the widely-used cloud-native tools. On-prem incumbents (Tabnine, Sourcegraph Cody Enterprise) are real and self-hostable — addressed in the cards below; none pair on-prem with AMD MI300X + CUDA→ROCm.

CAPABILITY	GITHUB COPILOT	CURSOR	CONTINUE.DEV	REPOMIND V3	CODEIUM ENT.
100% on-premises deploy	×	×	Partial	✓	Partial
256K token context (vLLM on AMD)	×	×	×	✓	×
AMD MI300X 192GB HBM3 support	×	×	×	✓	×
Automated CUDA → ROCm migration	×	×	×	✓	×
OWASP-LLM heuristic pre-scan (5 surfaces)	×	×	×	✓	Partial
EU AI Act Art. 12 record-keeping indicator	×	×	×	✓	×
docker compose up in 3 commands	×	×	Partial	✓	×
Gemma 27B cost-routing layer	×	×	×	✓	×

THE REAL INSIGHT

We don't compete with Copilot. We serve the orgs **that already restricted it** — and self-host incumbents (Tabnine, Cody Enterprise) don't do AMD-native CUDA→ROCm.

HONEST LANDSCAPE

On-prem coding AI exists (Tabnine, Cody Enterprise, Continue.dev). **None combine air-gap + MI300X 256K + CUDA→ROCm + a compliance pre-scan.**

THE WEDGE

The AMD MI300X + enforced air-gap combination is the differentiator. Not a permanent moat — a first-mover wedge into an underserved, forcing-function-driven segment.

Three commands. Enterprise-grade AI.

```
# Step 1 - Clone
git clone github.com/SRKRZ23/repomind-v3
cd repomind-v3

# Step 2 - Configure
cp .env.example .env
# FIREWORKS_API_KEY (smart routing, optional)
# VLLM_BASE_URL (AMD MI300X endpoint)
# ROUTING_THRESHOLD=0 -> zero-egress mode

# Step 3 - Launch
docker compose up

# -> http://localhost:3000
```

3 COMMANDS	256K VLLM CONTEXT	\$4.12 ACT 1 TOTAL	0 EGRESS (AIR_GAP=1)
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CODEBASE ANALYSIS

Multi-repo context (file tools scoped to the primary repo). SC-TIR (Structured Chain of Tool-Integrated Reasoning) agent loop: PLAN -> CALL -> OBSERVE -> THINK -> ANSWER — every step streamed live and persisted to an append-only audit log.

CUDA -> ROCM MIGRATION

Agent identifies CUDA files, applies 39 HIP rules + kernel-launch rewrite, generates an applicable unified diff. Complements AMD HIPIFY for common patterns.

SECURITY & COMPLIANCE SCAN

OWASP LLM security pattern detection + EU AI Act Article 12 audit trail check. AI-interpreted findings with line-level citations. Under 60 seconds for most repos.

Modes: (A) Smart routing — short -> Gemma 27B (Fireworks); long -> MI300X. (B) AIR_GAP=1 — enforced on-prem, Fireworks egress hard-refused. Model-agnostic: swap a Western/open-weight coder (Llama, Mistral) via VLLM_MODEL.

Fits one MI300X. Not one **H100**.

SINGLE NVIDIA H100 SXM 80GB — CAN'T HOLD IT

Model weights (FP8)		KV cache @ 256K ctx		Required VRAM
~77 GB	+	~95 GB	=	~172 GB

Single H100 VRAM: 80 GB → **single-card OOM. Needs multi-GPU sharding.**

2.4x

More VRAM than H100

256K

Token context window (vLLM config)

AMD MI300X 192GB HBM3 — FITS WITH HEADROOM

Weights		KV cache		Required		MI300X VRAM
~77 GB	+	~95 GB	=	~172 GB	<	192 GB ✓

Single GPU. No tensor parallelism. **+20 GB headroom.**

MODEL · VERIFIED ACT I

Qwen3-Coder-Next-FP8 · 80B/3B-active MoE
vLLM · AMD ROCm · max-model-len 262144
77.29 GiB weights · 94.58 GiB KV pool · 176/192 peak
\$4.12 · 124 min · Zenodo 10.5281/zenodo.20330468

Runs across the AMD line: **this ACT II live demo runs on AMD-hosted Fireworks** — every model call is AMD compute — while the 48GB Radeon / MI300X is the **on-prem tier**; the 256K / MI300X figures above are the **verified production tier** (ACT I, real MI300X). **100% AMD · open-weight models only (Gemma / GLM / Qwen) — no OpenAI/Anthropic/Gemini.**

Cheapest model that can do the job. Automatically.

```
// tiktoken cl100k_base - token counter
```

User Query

↓

```
[ count_tokens(context) ]
```

↓

```
ROUTING_THRESHOLD
| (default: 8,000 tokens, env var) |
|                                   |
```

BELOW threshold

ABOVE threshold

↓

↓

Gemma 27B

AMD MI300X

Fireworks AI

Qwen3-Coder-Next-FP8

~2s · \$0.001/q

15-90s · \$0.008-0.05+/q

```
// deep path: Gemma triage selects files →
```

```
// MI300X reasons over the focused subset
```

```
// AIR_GAP=1 → MI300X only (trriage off)
```

GEMMA 27B · SHORT QUERIES

Model: accounts/fireworks/models/gemma3-27b-it

Cost: \$0.0009 / 1K tokens · ~\$0.001/query

Latency: 1-3 seconds

Role: (1) fast tier for sub-8K queries; (2) **first-pass triage** — selects the relevant files so the deep pass isn't fed the whole repo (measured token reduction). Disabled under AIR_GAP=1.

On-prem design: the open-weight router (Gemma-class) runs on the enterprise's **own AMD GPU**, delegating deep reasoning to AMD-hosted Fireworks. In this ACT II demo **both tiers run on AMD-hosted**

Fireworks (Radeon notebook time-boxed) — fewer paid tokens = the enterprise cost-down pattern.

Prize track: Best Use of Gemma

AMD MI300X · DEEP ANALYSIS

Model: Qwen3-Coder-Next-FP8 (80B/3B-active MoE)

Cost: \$0.00028 / 1K tokens

Latency: 15-90 seconds

Context: 256K tokens (max-model-len 262144, verified)

SC-TIR AGENT

Structured Chain of Tool-Integrated Reasoning. Every query runs through PLAN → CALL → OBSERVE → THINK → ANSWER, streaming each step live to the UI.

Not a chatbot. A reasoning agent. Every step visible.

```
// Max 6 tool calls · 8 loop iterations
// Every step streamed as SSE event to UI
```

PLAN: Analyze authentication module for
 LLM06 (Sensitive Info Disclosure).

CALL: list_files("src/auth")

OBSERVE: src/auth/jwt.py
 src/auth/middleware.py

THINK: jwt.py is the risk surface.
 Grep for hardcoded secrets.

CALL: grep_codebase("SECRET_KEY")

OBSERVE: Line 12: SECRET_KEY="sk-prod-..." 🔍

ANSWER: LLM06 confirmed – Line 12.
 Remediation: use os.getenv().

PLAN

Transparent strategy before acting. Auditors see the reasoning chain — required for explainability compliance.

CALL

6 tools: read_file · grep_codebase · list_files · git_log · migrate_cuda_to_rocm · compliance_scan

OBSERVE

Tool results injected into context. Agent works on real code data — no hallucinated line numbers.

THINK

Explicit chain-of-thought before next action. Streamed to UI — every inference visible.

THE AI FLIGHT RECORDER

Every SC-TIR step is persisted to the **AI Flight Recorder** — a **tamper-evident, SHA-256 hash-chained + Ed25519-signed** append-only log. Any incident replays bit-exact; third-party verifiable, auditor-legible, aligned to EU AI Act Article 12.

NVIDIA built a **\$3T CUDA** moat. We automated the bridge.

```
--- original.cu
+++ converted.hip.cpp
@@ 39 rules + kernel-launch rewrite @@

-#include <cuda_runtime.h>
+#include <hip/hip_runtime.h>

- cudaMalloc(&d_A, bytes);
+ hipMalloc(&d_A, bytes);

- cudaMemcpy(d_A, h_A, cudaMemcpyHostToDevice);
+ hipMemcpy(d_A, h_A, hipMemcpyHostToDevice);

- cudaEvent_t start, stop;
+ hipEvent_t start, stop;

- cudaStreamCreate(&stream);
+ hipStreamCreate(&stream);

- cudaDeviceSynchronize();
+ hipDeviceSynchronize();

- matmul<<<grid, block, 0, stream>>>(...);
+ hipLaunchKernelGGL(matmul, grid, block, 0, stream, ...);

# 39 rules + kernel-launch. difflib unified diff (git apply-clean).
# cuBLAS/cuDNN calls, Thrust, PTX → roadmap (AI-guided).
```

39

CUDA→HIP rules + kernel-launch rewrite

HOW IT WORKS

The agent finds CUDA files and calls the migration tool, which applies 39 API-mapping rules (memory/stream/event/device/error + math-library headers) **and rewrites kernel-launch syntax** (kernel<<<grid,block>>> → hipLaunchKernelGGL), emitting an **apliable unified diff** (git apply-clean).

Unit-tested against demo/cuda_sample.cu. Complements AMD HIPIFY.

WHY AMD WINS THIS

50M+ lines of CUDA code on GitHub. The #1 barrier to AMD GPU adoption is migration friction. REPOMIND removes it automatically. **Roadmap: AI-guided migration for non-trivial CUDA patterns (custom kernels, PTX).**

A heuristic pre-scan in **under 60 seconds**. Flags surfaces for human review — not a certified control.

REPOMIND HEURISTIC PRE-SCAN – enterprise-repo
NOT a certified control. Flags for human review.

LLM01 Prompt Injection surfaces: – no match
LLM02 Insecure Output: ▲ REVIEW
 ui/render.ts:47: innerHTML = ...
LLM06 Sensitive Info Disclosure:▲ REVIEW
 auth/jwt.py:12: api_key = "sk-prod-..."
LLM07 Unvalidated Outbound Call: – no match
LLM10 Untrusted Deserialization:– no match

EU AI Act Art. 12 indicator:
 logging/audit API usage detected

Findings cite real file:line. ~45s.
Word-boundary matching (no substring noise).
Hash-chained + Ed25519-signed (verify_signatures)

5 OWASP-LLM RISK SURFACES (HEURISTIC)

Word-boundary detection across LLM01, LLM02, LLM06, LLM07, LLM10 with real file:line citations. A heuristic pre-scan, **not a certified control** — a clear result is not an assurance. Extensible to full Top 10 (roadmap).

EU AI ACT ARTICLE 12

Detects presence/absence of audit logging in your AI system codebase. Article 12 requires audit trails for AI systems — REPOMIND flags the gap before your compliance team does. **Note: applicability depends on AI system risk classification.**

SHIPPED: AI FLIGHT RECORDER

Shipped: the **AI Flight Recorder** — Ed25519-signed, hash-chained audit records. Regulators/auditors verify each record's authenticity with the public key (verify_signatures) — an immutable, third-party-verifiable, replayable evidence chain, not just plain text.

ACT I winner. CVPR co-author. Building in public.

AMD HACKATHON ACT I — 1ST PLACE

1st Place AI Agents + Outstanding Social Engagement

Prize: AMD Radeon™ AI PRO R9700 + cash. AMD published official developer testimonial naming REPOMIND + RZS. 124-minute benchmark, 62 data points, \$4.12 total. Zenodo DOI 10.5281/zenodo.20330468. **Reproducible — every step documented.**

3 ZENODO PREPRINTS · ORCID VERIFIED

REPOMIND v1 — DOI 10.5281/zenodo.20330468

TriageGuard (Healthcare AI) — DOI 10.5281/zenodo.20445507

ECB LLM paper — DOI 10.5281/zenodo.19791329

ORCID: 0009-0007-0731-4247

Apache 2.0 · MIT · All reproducible.

CVPR 2026 WORKSHOP — CO-AUTHOR

"The First AI Children Gait Challenge"

CVPR 2026 Workshop Proceedings. Public Kaggle LB: #2 / 56 teams. Official post-evaluation rank: 6/7 (formula changed 0.5/0.5 → 0.7S1+0.3S2 at eval). **Co-author credit on proceedings paper confirmed regardless of final rank.**

THE PATTERN

Solo. Uzbekistan. Winning global competitions. Building real infrastructure.

Not a team of 20. One developer, verified benchmarks, public preprints, open source. The trajectory: hackathon → CVPR → enterprise infrastructure. **Each step reproducible.**

The math is simple. The regulatory risk is not.

JP MORGAN CHASE

~\$1–2B

Recoverable productivity / yr (est.)

50,000 devs · McKinsey ~37% is a **task-level** speedup on coding tasks (~20–40% of dev time), not 37% of total comp → recoverable value ≈ **\$1–2B/yr**

REPOMIND on-prem license: **\$0.5–2M/yr**
Payback: **months, after a 6–18mo sales cycle**

APPLE INC.

\$1B+

IP leak exposure per incident

30,000+ engineers · \$200–400K/yr
IP leak: **\$100M–\$1B per event**
REPOMIND: zero egress (MI300X mode)

All code stays in Apple's datacenter.
Risk reduction: eliminates surface entirely.

GOLDMAN + EU ENTERPRISES

€7.5–35M

EU AI Act penalty tiers²

EU AI Act obligations phase in **through 2026**
Public statements on immature AI governance¹
REPOMIND: Art. 12 record-keeping *indicator*

A record-keeping gap sits in the **≤€7.5M tier**
Applicability depends on AI-system risk classification.

Value at risk across the segment is large (billions in recoverable productivity + tiered regulatory exposure + IP-leak risk). **Serviceable revenue is a fraction of that** — see the revenue model. **Few combine on-prem + AMD-native today.**

Open source core. Enterprise features commercial.

STARTER \$500 /month ✓ 1 team, up to 10 users ✓ Public repos only ✓ Gemma 27B routing ✓ OWASP pattern scan ✗ On-prem MI300X ✗ Signed reports	PROFESSIONAL \$2,500 /month ✓ 25 users ✓ Private repos ✓ On-prem Docker deploy ✓ Compliance reports ✓ Priority support ✗ Air-gapped mode	ENTERPRISE \$15K /month ✓ Unlimited users ✓ Air-gapped MI300X mode ✓ CUDA→ROCm migration ✓ Ed25519-signed audit log ✓ SLA + dedicated support ✓ Custom procurement	AMD CLOUD Usage -based ✓ Pay-per-query ✓ \$0.008–0.05+/deep query ✓ \$0.001/short query ✓ AMD Developer Cloud ✓ No infra overhead ✓ Global availability
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WHY ENTERPRISE BUYS Demand is driven by productivity recovery + risk reduction. Even a modest slice of the \$1–2B recoverable productivity dwarfs a \$180K/yr license. Regulatory pressure adds urgency (tiered penalties²).	OPEN SOURCE CORE STRATEGY Core agent + tools MIT-licensed. Enterprise features (air-gap mode, signed reports, SLA) commercial. Classic Elastic/HashiCorp model — community builds trust, enterprise pays.	ACT I TRACTION = FIRST SIGNAL AMD official testimonial + 1st place = validated by hardware vendor. Next milestone: one enterprise pilot conversation post-hackathon. Target: Q3 2026.
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REPOMIND is the **trust + cost layer** for on-prem enterprise AI on AMD.

Jul 11 ACT II SUBMIT	Jul 12 SHOW HN LAUNCH	Q3 2026 FIRST PILOT	Q4 2026 SOC2 PATH + SLA
TIER 1 · CORE — SHIPPED (ACT II) ✓ On-prem coding agent on AMD (Radeon → MI300X) ✓ Gemma-local → Fireworks routing + measured cost layer ✓ AI Flight Recorder (Ed25519 tamper-evident) ✓ CUDA→ROCm · air-gap · compliance pre-scan skeleton ~89% · cache: repeat ctx ~90% input · 38 tests · Docker	TIER 2 · TRUST & GOVERNANCE — NEAR-TERM → Cost & Savings FinOps panel (live \$ saved) → Self-verifying answers (local verifier) → PII redaction gateway (pre-egress) → Continuous on-prem red-team → Sovereign RAG over regulated corpora	TIER 3 · SOVEREIGN & REGULATED — HORIZON → Sovereign AI-in-a-box for regulators → Verifiable unlearning (GDPR Art. 17) → AI-decision insurance / underwriting → Carbon + cost-aware routing → Trusted AI metering standard	

Every tier runs on the same spine: **cheap open-weight compute on the customer's own AMD GPU on-prem, bursting to AMD-hosted Fireworks only when it pays to** — the economics that make the whole platform defensible. **Tier 1 is shipped; Tiers 2–3 are roadmap, not claims.**

DEPLOY IT. RIGHT NOW.

REPO**MIND**

v3

```
git clone github.com/SRKRZ23/repomind-v3
cp .env.example .env
docker compose up
```

GITHUB.COM/SRKRZ23/REPOMIND-V3

AMD ACT II · TRACK 3 UNICORN

ACT I 1ST PLACE

GEMMA 27B · FIREWORKS AI

MIT LICENSE · OPEN SOURCE

\$500–\$15K

Monthly enterprise tiers

\$1–2B

Recoverable productivity / yr (est.)

Aug 2026

EU AI Act enforcement

Built by **Sardor Razikov** · Solo researcher, Tashkent, Uzbekistan

CVPR 2026 Workshop Co-Author · ORCID 0009-0007-0731-4247 · 3 Zenodo preprints · AMD ACT I 1st Place